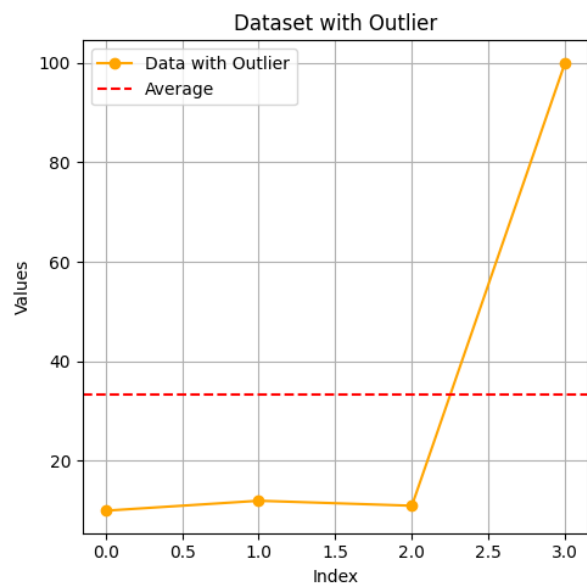
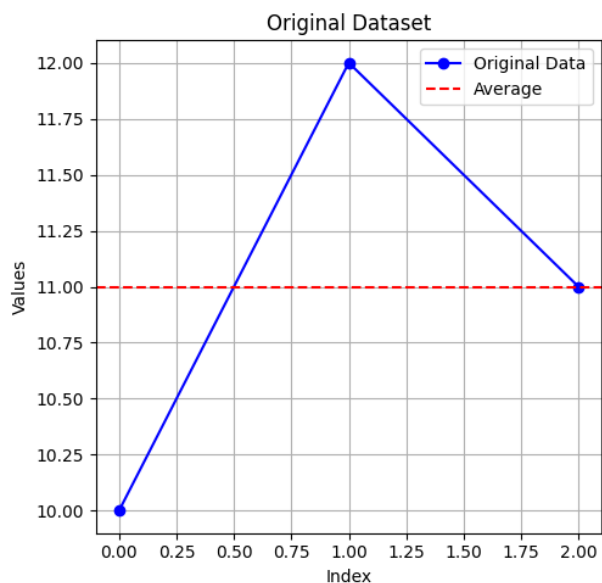


## Outliers

- In a dataset, some data points are very different from the rest of the whole dataset.
- Dataset: 10, 12, 11, 13, 100.
- 100 is outlier cuz it is different from rest of the data points.
- outliers can be caused by wrong or mismatched data collection.
- It is essential to find out and handle outliers.
- Average shift” means the analysis result also shifts (changes).
- They give you good and bad values
- **Small example:**
- Dataset: 10, 12, 11  
Average = **11**  
Add an outlier: 10, 12, 11, 100  
New average = **33.25**
  - the **prediction or analysis goes in the wrong direction** because of the outlier.
  - Outliers can **mislead models and analysis** if not handled properly.



Mean (Average)

**Formula:**

$\text{Mean}(\mu) = \text{Number of values} / \text{Sum of all values}$

**Example:** Dataset = [10, 12, 11]

1. Add all values:  $10 + 12 + 11 = 33$
2. Divide by number of values:  $33 \div 3 = 11$
3. **Mean = 11**

## Variance

Variance shows how spread out the data is from the mean.

**How outliers show errors or anomalies:**

1. **Look at big differences**
  - If one number is much bigger or smaller than the rest → might be wrong.
  - Example: [10, 12, 11, 100] → 100 looks unusual
2. **Compare with average**
  - If a number is far away from the average → possible error

**Types of Outliers:**

## Univariate Outliers

- “Uni” = one variable
- Outlier in a **single column/feature**
- Example: Heights of people = [150, 160, 155, 300] → **300** is an outlier in height

## Multivariate Outliers

- “Multi” = more than one variable
- A data point may look normal in one feature, but weird when looking at multiple features together
- Example:
  - Height = 170 cm, Weight = 50 kg → seems fine individually
  - But for a 170 cm adult, 50 kg is unusually light → multivariate outlier

## Global Outlier

Global (Point) Outlier means:

- One single data point that is very different from all the other points in the dataset.

- It stands alone far away from the rest.

### **Contextual Outlier**

- A data point that is **unusual in a specific context**, but may look normal if you ignore the context.
- **Depends on surrounding conditions** (like time, season, location, etc.)

### **Example 1: Temperature**

- Winter: 25°C → **unusual** → contextual outlier
- Summer: 25°C → **normal** → not an outlier

### **How to identify outliers**

#### **1. Visually identify means create a plot**

- Boxplot
- Histogram
- Z score method